

Colors are determined by different combinations of chemicals.

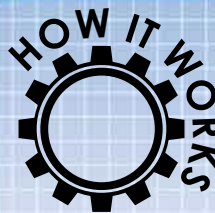


Gunpowder plot

In the Middle Ages, rockets became **weapons**, catapulted into enemy bases to explode on impact and set camps on fire. Famous English conspirator **GUY FAWKES** plotted a similarly dastardly deed with gunpowder. His failed attempt to **blow up** London's Houses of Parliament on November 5, 1605, is remembered every year with dummies of Fawkes **BURNED AT FIREWORK DISPLAYS**.

Spectrum of color

The future is **glowing bright** for fireworks. Color chemistry has developed a rainbow effect—magnesium and aluminum make **white** light, sodium salts make **yellow**, copper salts make **blue**, strontium nitrate or carbonate makes **red**, and barium nitrate makes **green**. Scientists are researching different **CHEMICAL COMBINATIONS** in order to make ever more impressive fireworks.



Once lit, the fuse allows you to retreat to a safe distance before the fuel propellant sends the firework into the sky. Once there, a part of the fuse called the delay controls when the chemical reaction in the rocket happens by slowing the ignition of the gunpowder. When the gunpowder is lit, it creates loud banging sounds and shoots the stars out of the firework.



How it changed

Fireworks have delighted audiences for centuries, but the gunpowder at their core has also been used as a devastating weapon of war.

the world



The strong, stable explosive **DYNAMITE** was developed from gunpowder. Used in **CONSTRUCTION**, it is also a harmful **WEAPON**.



Key dates are celebrated with **FIREWORK DISPLAYS**. On New Year's Eve, many countries set off fireworks at midnight to **WELCOME THE COMING YEAR**.